

Boundary-Blind World Models: Why JEPA Predictors Smear Contact Bifurcations, and What It Takes to Fix Them

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Abstract

Action-conditioned JEPA world models plan well in free space and hit a contact wall. We diagnose the wall directly on DINO-WM and JEPA-WM in Metaworld, then test whether it disappears under DROID encoder scaling up to V-JEPA-2-AC. The failure is not a generic accuracy gap but a structural failure we call *Boundary Blindness* (BB): near contact boundaries — where small action changes produce qualitatively different futures — unimodal latent predictors trained with full-latent L_2 collapse the outcome fan to its mean. We (1) define a frozen-checkpoint diagnostic that measures BB against ground-truth outcome spread and show it concentrates at the pre-grasp boundary on two Metaworld model families (BB 1.32/1.28 at the boundary vs. 0.28/0.30 in free space), transfers to real-robot DROID data, and persists in DROID diagnostic scaling across a $45\times$ encoder scale-up (22M to 1.01B, DINOv2/v3 and V-JEPA-2) that leaves action-grounding at the chance floor; (2) tie BB to Metaworld planning: at upstream-parity protocol our harness *reproduces* the published Metaworld Reach baseline (episode-end 37.5% [18.5–61.4] vs. reported 44.8 ± 8.9 — inside the CI) yet both the L_2 and grounded planners score 0/16 on push and pick-and-place — the arm reaches the goal pose (2–4 cm) while the object never moves, the closed-loop signature of BB; (3) localize the failure by ablation: mixture predictor heads and metric re-weighting are quantified nulls, while a 0.5M-parameter grounded dynamics channel $h(z, a) \rightarrow \Delta \text{obj}$ trained on the same Metaworld cached data restores counterfactual tracking (corr $0.03 \rightarrow 0.68$) and halves boundary BB on both models, improving the closed-loop cost surface (paired distance $+0.089$, CI $[+0.022, +0.162]$) without yet flipping task success. The bottleneck is the training target, not the data, the capacity, or the metric.

1 Introduction

Action-conditioned world models that predict in a learned latent space — the JEPA family (DINO-WM (Zhou et al., 2024), V-JEPA-2-AC (Assran et al., 2025), and the design study of (jepa-success authors) (2025)) — have become an attractive substrate for robot planning: they reuse strong self-supervised visual encoders, predict cheaply in representation space, and plan zero-shot from a goal image with sampling-based model-predictive control. On free-space tasks they work well. But a striking pattern runs through the field’s own evaluations: the *positive* closed-loop numbers are reported for reaching and pose control, and the contact-rich tasks — pushing, grasping, pick-and-place — are either omitted, demoted to qualitative figures, or propped up with hand-engineered sub-goals. (jepa-success authors) (2025) stop their closed-loop Metaworld tables at Reach and Reach-Wall and note in the appendix that on object-manipulation tasks the model “hallucinates grasping the object”. Assran et al. (2025) report pick-and-place only after supplying three hand-crafted sub-goal images that walk the planner through the grasp. The wall is real, and it is at contact.

We argue — and then show across a $45\times$ encoder scale-up (Sec. 3.5) — that this is not a generic accuracy deficit a bigger encoder would erase, but a *structural* failure of the prediction objective. Contact induces *outcome bifurcations*: from nearly the same state, a small change in action produces qualitatively different futures (the gripper centred over an object lifts it; two or three degrees off, it does not). A unimodal latent predictor trained with an L_2 (or cosine) reconstruction loss is, by construction, a conditional-mean estimator — it averages across the outcome fan rather than representing its branches. Crucially, the averaging is invisible to the metrics one would naturally reach for. The one-step prediction z_{t+1} after a small action perturbation is near-continuous; the divergence is a property of the *rollout* and of the *outcome*, not of any

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single transition. A diagnostic that asks “does the model use actions?” will answer “yes” even as the model is blind to exactly the action distinctions that decide a grasp.

We make this precise. First we show that the action-grounding the field already measures is intact in the easy direction and broken in the hard one: frozen DINO-WM and JEPA-WM rank a factual action against its *opposite* near-perfectly everywhere (counterfactual ranking accuracy 0.96–0.99), yet collapse when the distractor is a *nearby* action (0.47–0.57), and most so in the pre-grasp regime. To turn this from a symptom into a measurement we introduce *Boundary Blindness* (BB): on a neighbourhood of similar states where the *world’s* true outcomes fan out, BB is the rectified gap between that ground-truth action-sensitivity and the model’s. Because BB references the world, a high value cannot be explained away by small one-step deltas. On frozen Metaworld checkpoints BB concentrates sharply at the pre-grasp boundary (1.32 vs. 0.28 in free space) on both DINO-WM and JEPA-WM and transfers to real-robot DROID data. As a DROID diagnostic, it is also *scale-invariant*: across a $45\times$ encoder scale-up and two self-supervised families (DINOv2 ViT-S 22M, DINOv3 ViT-L 300M, V-JEPA-2-AC ViT-G 1.01B, and an independent public V-JEPA-2-AC), action-grounding stays at the chance floor and BB stays positive — ruling out “just scale the encoder”.

We then close the loop in two senses. We connect BB to *planning*: at a protocol matched to the upstream evaluation, planning succeeds exactly where BB is low — our harness reproduces the published Metaworld Reach baseline (the L_2 planner scores episode-end 37.5% [18.5–61.4], inside the reported 44.8 ± 8.9) — and collapses where BB is high, with the signature BB predicts: on push and pick-and-place both the baseline and an improved planner score 0/16, the end-effector reaching the goal pose (2–4 cm) while the object never moves. Finally we ask *why* the model is boundary-blind, with a controlled fix ladder on frozen trunks. Adding capacity (mixture-density predictor heads) does not move BB; re-weighting the latent metric does not move it; a probe chain shows the object state is decodable from the latent and propagated for factual actions, but the *counterfactual* action→object channel is noise. A 0.5M-parameter grounded dynamics channel $h(z, a) \rightarrow \Delta\text{object}$, trained on the same cached data with everything else frozen, repairs exactly that channel: counterfactual tracking rises from 0.03 to 0.68, boundary BB halves on both models, and the closed-loop planning cost surface improves measurably (paired final-distance gain +0.089, CI [+0.022, +0.162]) — though not yet enough to flip task success, which additionally requires contact-aware action proposal. The diagnosis is the point: the bottleneck is the training *target*, not the data, the capacity, or the metric.

Our contributions:

- **C1 (diagnostic: CounterfactualBench + Boundary Blindness).** A frozen-checkpoint diagnostic suite. Existing action-grounding metrics (Counterfactual Ranking Accuracy CRA, Action Usage Gap AUG, Effect-Conditional Sensitivity ECS) give the first signal: the models rank a factual action against its *opposite* near-perfectly everywhere (CRA 0.96–0.99) but collapse against *nearest-neighbour* action distractors, worst at the pre-grasp boundary (CRA 0.47–0.57). *Boundary Blindness* (BB), our new metric, explains why and localizes it: the rectified gap between the world’s local action-sensitivity and the model’s, on similar-state neighbourhoods. BB concentrates at the pre-grasp boundary on both DINO-WM and JEPA-WM (CI-aware, per task), transfers to DROID, and remains present in the DROID scaling diagnostic through V-JEPA-2-AC (Sec. 3).
- **C2 (planning link).** At upstream-parity Metaworld closed-loop protocol, planning succeeds exactly where BB is low (Reach: L_2 37.5%, grounded 50.0%, inside the published 44.8 ± 8.9 CI) and collapses where BB is high (Push, Pick-and-place: 0% both arms) with the signature BB predicts: the end-effector arrives, the object never moves (Sec. 4).
- **C3 (mechanism + fix).** A controlled fix ladder on frozen trunks: mixture heads — null; metric re-weighting — null; a probe chain showing the object state is *present* in the latent and propagated for factual actions but dead for counterfactual ones; and a 0.5M-parameter grounded dynamics channel that repairs exactly that channel on both model families (Sec. 5).

2 Related work

JEPA world models for planning. DINO-WM (Zhou et al., 2024) predicts forward in the frozen feature space of a self-supervised image encoder and plans with sampling-based MPC from a goal image; V-JEPA-2-AC (Assran et al., 2025) learns an action-conditioned world model on top of a video encoder

and demonstrates zero-shot manipulation; and (jepa-success authors) (2025) systematically study the design choices (encoder, action conditioning, rollout loss, planner) that make such models plan well. We take these as our frozen baselines and treat their evaluations as evidence: across all three, the strong closed-loop results are pose/free-space tasks, and the contact tasks are where the story thins. (jepa-success authors) (2025) restrict their closed-loop Metaworld tables to Reach and Reach-Wall and report qualitatively that object-manipulation rollouts “hallucinate grasping”. Assran et al. (2025) achieve pick-and-place only with three hand-specified sub-goal images (and grasp-from-goal-image success of 25–65%), while their behaviour-cloning baseline Octo (Octo Model Team et al., 2024) grasps at 0–20%. Our work gives this shared wall a name, a metric, and a mechanism.

Multimodal and distributional dynamics. That a deterministic next-step predictor averages multimodal futures is a classic observation in video prediction and model-based RL; mixture-density and latent stochastic predictors are the standard response. A contribution of our fix ladder is the *negative* result that this standard response does not apply here: adding a mixture-density head over the latent leaves Boundary Blindness unmoved (Sec. 5), because the bifurcation occupies a tiny, under-weighted subspace of the L_2 geometry and the boundary’s action-dependence is not recoverable from expert-only data at the head level. The fix is to ground a small dynamics channel in an explicit outcome variable, not to make the latent predictor distributional.

Diagnosing action-grounding. Prior diagnostics ask whether a world model uses its action input at all — permuting or zeroing actions and measuring the change in prediction error. Our CounterfactualBench (CRA/AUG/ECS) includes such measures and confirms they are necessary but not sufficient: they are satisfied (the model clearly uses actions) precisely where planning still fails. Boundary Blindness is, to our knowledge, the first diagnostic to compare a model’s local action-sensitivity against the *world’s* ground-truth sensitivity on a matched neighbourhood, which is what lets it isolate the contact boundary rather than conflate it with low-motion transitions.

Metaworld and DROID. We evaluate on Metaworld (Yu et al., 2020), whose simulator state lets us define and supervise an object-displacement outcome label, and use DROID (Khazatsky et al., 2024) as a real-robot transfer check (proprioceptive $\|\Delta z\|$ proxy only — no object ground truth). Planning follows the CEM protocol of Rubinstein (1997) as configured by the upstream evaluation.

3 Diagnosis: CounterfactualBench and Boundary Blindness

We run a frozen-checkpoint diagnostic suite (nothing is trained). It has two layers: existing action-grounding metrics that establish *that* the models fail to discriminate nearby actions in contact regimes, and our new Boundary Blindness metric that explains *why* and pins the failure to the contact boundary. We then show the failure is robust — it transfers to real-robot data and persists unchanged across a $45\times$ encoder scale-up (Sec. 3.5).

3.1 CounterfactualBench: action-grounding metrics

Let $F(z, a)$ be the frozen one-step latent predictor (driven through the model’s own `unroll` primitive, with the model’s action normalisation) and $d(\cdot, \cdot)$ the planning distance (L_2 for every baseline, matching the upstream `L2_cem` configuration). We measure three quantities on factual transitions (z_t, a_t, z_{t+1}) .

CRA (Counterfactual Ranking Accuracy). The factual prediction should be closer to the realised next state than predictions under K distractor actions:

$$\text{CRA} = \Pr \left[d(F(z_t, a_t), z_{t+1}) < \min_{k=1\dots K} d(F(z_t, a_k^-), z_{t+1}) \right]. \quad (1)$$

We report top-1 accuracy under two distractor families: *opposite* ($a^- = -a_t$, easy) and *nearest-neighbour* (a^- drawn from the closest other executed actions in the similar-state neighbourhood of Sec. 3.2, hard).

AUG (Action Usage Gap). How much does the realised action help over a random one,

$$\text{AUG} = \mathbb{E} \left[d(F(z_t, \pi(a_t)), z_{t+1}) - d(F(z_t, a_t), z_{t+1}) \right], \quad (2)$$

with π a batch permutation of actions; $\text{AUG} \approx 0$ iff actions are ignored. (AUG is a raw latent gap, comparable across regimes within a model but not across models with different latent scales; CRA, a ranking, is the cross-model-comparable metric.)

ECS (Effect-Conditional Sensitivity). AUG, and the primary *effect-conditioned* CRA, restricted to transitions that actually move the latent, $\|z_{t+1} - z_t\| > \tau$ with τ the per-model median displacement. This removes near-static steps where the action genuinely should not change much, so a low effect-conditioned score reflects true action-ignoring rather than a quiet transition. Confidence intervals are trajectory-clustered bootstrap. We report AUG and ECS in full in App. B (both collapse toward zero, corroborating the CRA result); the main text leads with CRA because, as a ranking, it is the cross-model-comparable metric.

Table 1: Effect-conditioned CRA (top-1, n_{eff} -weighted pool over tasks). Against *opposite* distractors the models are near-perfect everywhere — they *can* use actions. Against *nearest-neighbour* distractors CRA collapses, worst at the pre-grasp boundary: the models cannot tell apart *nearby* actions exactly where it matters. This motivates Boundary Blindness.

model	distractor	free space	pre-grasp	contact
DINO-WM	opposite	0.990	0.961	0.975
	nearest-neighbour	0.566	0.467	0.481
JEPA-WM	opposite	0.993	0.972	0.984
	nearest-neighbour	0.635	0.541	0.571

The opposite-vs-nearest-neighbour split is the key precursor: the models are not action-blind in general (opposite CRA ≈ 0.97 – 0.99); they fail specifically to resolve *small* action differences, and most in the pre-grasp/contact regimes. CRA alone cannot say whether this is a real boundary failure or just small one-step latent deltas — which is what BB resolves.

3.2 Boundary Blindness: definition

The idea is to compare, on a neighbourhood of similar states, how much the *world’s* outcome fans out under nearby actions against how much the *model’s* prediction fans out under the same actions. Where the world bifurcates but the model does not, the model is blind to the boundary.

Neighbourhood. For each anchor transition i with latent state z_i and action a_i we form a similar-state neighbourhood $\mathcal{N}(i)$ from the cached reference set: its M ($= 16$) nearest transitions in latent distance $\|z_i - z_j\|$, keeping those within a radius ρ and, when fewer than M fall inside, relaxing to the M nearest (this is the same neighbourhood the hard-negative CRA sampler uses; the relaxation is necessary because a fixed radius in raw latent units otherwise selects almost nothing). Let o_j denote the *true* scalar outcome of neighbour j — object displacement $\|\Delta \text{obj}_j\|$ on Metaworld, the latent-change proxy $\|\Delta z_j\|$ on DROID — and let $\mathcal{M}(i)$ be the valid (in-radius or relaxed) members.

Boundary regime. A transition lies in the *boundary regime* when small action changes produce large true-outcome changes across its neighbourhood:

$$\text{boundary_score}_i = \frac{\text{std}_{j \in \mathcal{M}(i)} o_j}{\text{mean}_{j \in \mathcal{M}(i)} \|a_j - a_i\| + \varepsilon}. \quad (3)$$

A high score requires outcomes to fan out under nearly identical actions; note this is *not* the same as “large effect” — free-space drift has large $\|\Delta z\|$ but smooth, low-variance neighbourhoods and so scores low. A percentile threshold (mirroring the ECS effect mask) turns the score into the boundary mask used for $\text{BB}_{\text{boundary}}$.

The two sensitivities. The world’s local action-sensitivity at anchor i is the dispersion of true outcomes across the neighbourhood,

$$S_i^{\text{true}} = \sqrt{\frac{1}{|\mathcal{M}(i)|} \sum_{j \in \mathcal{M}(i)} (o_j - \bar{o}_i)^2}, \quad (4)$$

a property of the data alone. The model’s local action-sensitivity applies each neighbour’s action a_j to the *single* anchor state z_i and measures how far the resulting predictions spread, as the root-mean-square latent distance to their centroid $\bar{F}_i = \frac{1}{|\mathcal{M}(i)|} \sum_j F(z_i, a_j)$,

$$S_i^{\text{model}} = \sqrt{\frac{1}{|\mathcal{M}(i)|} \sum_{j \in \mathcal{M}(i)} \|F(z_i, a_j) - \bar{F}_i\|^2}, \quad (5)$$

a property of the frozen predictor F .

Boundary Blindness. Because S^{true} and S^{model} carry different units (outcome vs. latent), each is standardized (z-scored) over the whole reference population $\mathcal{P}$, $\tilde{S} = (S - \mu_{\mathcal{P}}(S)) / \sigma_{\mathcal{P}}(S)$, and Boundary Blindness is the rectified gap

$$\text{BB}_i = \text{relu}(\tilde{S}_i^{\text{true}} - \tilde{S}_i^{\text{model}}). \quad (6)$$

$\text{BB}_i \approx 0$ means the model’s action-sensitivity matches the world’s; BB_i large means the world bifurcates at i while the model predicts nearly the same future for every nearby action — it is blind to the boundary. $\text{BB}_{\text{boundary}}$ pools BB over the boundary-masked transitions. Standardizing over the population (not per anchor) keeps BB comparable across regimes, which is what lets us read off “BB is high at pre-grasp, low in free space”. Unlike CRA, BB references the world, so a high value cannot be an artefact of small one-step latent deltas. All reported BB values are pooled n_b -weighted with trajectory-clustered bootstrap confidence intervals.

3.3 Setup

We operate entirely on pretrained, frozen checkpoints (DINO-WM and JEPA-WM for Metaworld; four world models from DINO-WM 22M to V-JEPA-2-AC 1B for DROID, see Sec. 3.5), loaded through the official hub interface and driven through the model’s own `encode/unroll` primitives — nothing is trained for the diagnostic. Every frame is encoded once and cached; all metrics run on the cache. We stratify transitions into four regimes — *free space*, *pre-grasp*, *gripper actuation*, and *contact manipulation* — using a proxy computed from the 39-dimensional Metaworld state (end-effector and object positions, with object displacement as the contact proxy, since the released dataset carries no MuJoCo contact ground truth); DROID uses proprioception and latent-change heuristics. Confidence intervals are trajectory-clustered bootstrap throughout. Two honesty caveats carry on every BB number: the boundary outcome label is an object-displacement *proxy*, and we exclude the articulated-object task mw-door-close, where the proxy is anomalous (we report it hatched as a sensitivity check).

3.4 Result: BB concentrates at the pre-grasp boundary

Table 2: Pooled $\text{BB}_{\text{boundary}}$ (n_b -weighted, mw-door-close excluded) on frozen checkpoints. Per-task CI-aware pairing: pre-grasp elevated in 4/6 (DINO-WM) and 5/6 (JEPA-WM) tasks, zero confident reversals. DROID: $\|\Delta z\|$ proxy, transfer check only.

	free space	contact	pre-grasp boundary
DINO-WM (Metaworld)	0.282	0.481	1.323
JEPA-WM (Metaworld)	0.299	0.441	1.280
DINO-WM (DROID, $\ \Delta z\ $)	0.721 [0.613, 0.834]	—	1.975 [1.601, 2.350]

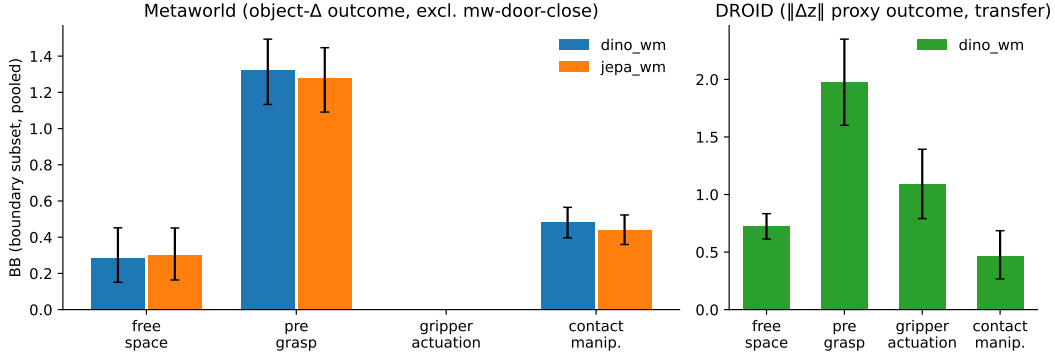


Figure 1: BB per regime, frozen baselines (Metaworld both families + DROID transfer).

3.5 The failure does not scale: a $45\times$, two-family curve

A natural objection is that action-grounding is simply a capacity problem that a larger encoder would solve. It is not. This section tests whether diagnostic action-grounding/BB failures disappear with encoder scale on DROID; it is not a closed-loop or grounded-fix result. We score four frozen DROID world models on the identical 333-episode / 2331-transition subset and gripper-primary regimes, spanning a **$45\times$ encoder scale-up and two self-supervised families**: DINO-WM (DINOv2 ViT-S/14, $\sim 22\text{M}$), JEPA-WM (DINOv3 ViT-L/16, $\sim 300\text{M}$), V-JEPA-2-AC (ViT-G/16, $\sim 1.01\text{B}$), and an independently trained public V-JEPA-2-AC checkpoint (V-JEPA-2-AC-OSS, ViT-G/16). We lead with the deterministic hard-negative metrics (nearest-neighbour distractor pool and bootstrap CIs are seeded and reproduce bit-identically; the easier *random/opposite* pools are unseeded and drift run-to-run, so we do not report them here).

Table 3: DROID action-grounding does not scale. *Top*: effect-conditioned CRA (top-1, *nearest-neighbour* distractors); 16-way chance = 0.0625. All four models sit *at the chance floor* in the action-critical regimes — the 300M DINOv3 is no better than the 22M DINOv2, and the 1B V-JEPA-2 models are if anything slightly lower on contact. *Bottom*: $\text{BB}_{\text{boundary}}$ is positive in every cell of every model (boundary-blind), with the pre-grasp locus in 3/4. Scaling the encoder $45\times$ and switching family closes neither gap.

regime	DINO-WM 22M	JEPA-WM 300M	V-JEPA2-AC 1B	V-JEPA2-AC-OSS 1B
<i>effect-CRA, nearest-neighbour (chance = 0.0625)</i>				
pre-grasp	0.068	0.077	0.061	0.069
gripper actuation	0.031	0.059	0.049	0.031
contact manipulation	0.055	0.045	0.035	0.033
$\text{BB}_{\text{boundary}} (> 0 = \text{boundary-blind})$				
pre-grasp	1.916	1.205	1.933	1.560
free space	0.963	1.103	1.010	0.819
gripper actuation	1.088	1.057	0.643	0.828
contact manipulation	0.421	0.897	0.390	0.595

The curve is **flat across $45\times$ scale**: effect-CRA against hard negatives stays at the 16-way chance floor in pre-grasp, gripper-actuation, and contact for all four models, and BB is positive (pre-grasp locus) in every cell. A capacity sanity check confirms the metric is not saturated low by construction — on toy fully-actuated datasets (Push-T, PointMaze, Wall) the same diagnostic returns effect-CRA 0.66–1.0, so the near-chance contact scores are real failures, not a broken metric. “Just scale the encoder” is ruled out as a fix; the failure is structural, consistent with the Boundary-Blindness account.

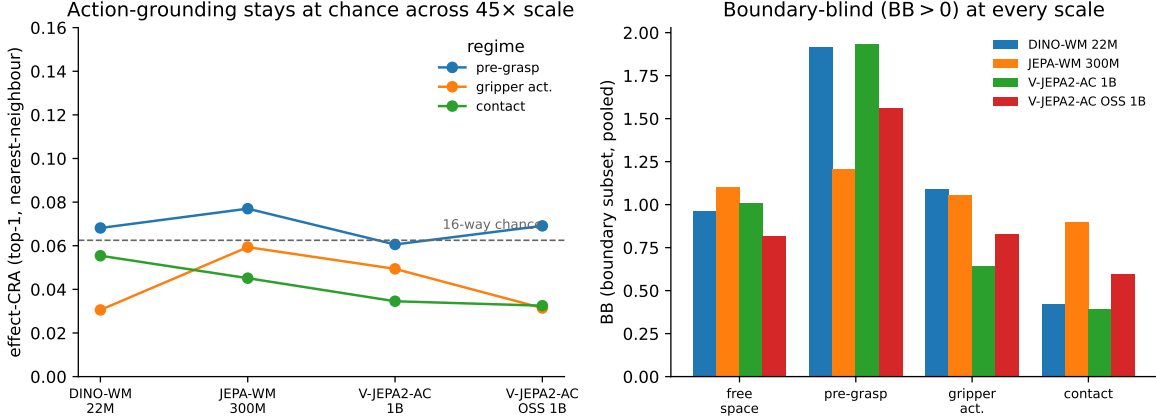


Figure 2: The DROID contact failure does not scale. *Left*: effect-conditioned CRA against nearest-neighbour distractors stays at the 16-way chance floor (dashed) across a 45× encoder scale-up in every action-critical regime. *Right*: BB_{boundary} is positive (boundary-blind) in every regime of every model, with the pre-grasp locus.

4 Boundary Blindness predicts closed-loop planning failure

4.1 Protocol

We match the upstream Metaworld evaluation, read off its shipped configuration: the goal is the scripted expert’s final frame, planning is CEM in latent space with horizon 6, 300 samples, 15 iterations, executing 3 model-steps per replan, for at most 100 environment steps, and success is the simulator’s flag. The planning cost is latent L_2 to the goal ($\alpha = 0$; proprioception remains in the predictor context, which the architecture requires). We run 16 paired episodes per task: within a pair the two arms (L_2 baseline and the grounded cost of Sec. 5) share the environment, its frozen initialisation, and the CEM noise seeds, so the comparison is exactly paired. Reaching a faithful protocol required fixing three environment-side reproduction bugs — a renderer that silently used the default camera at 480px, a missing vertical flip in the training pipeline, and a goal-frame convention — detailed with their verification in Appendix A; after the fixes the rendered observations are indistinguishable from training data to the model (one-step prediction error $1.4\times$ the dataset level, latent nearest-neighbour ratio 0.97), while the physics is identical (1.5 mm replay error).

4.2 Results

Table 4: Closed-loop Metaworld, 16 paired episodes per task. Success is judged at the end of the episode, matching the upstream convention (`--strict-success`; SLURM 22256). Paper baseline: DINO-WM CEM- L_2 MW-Reach 44.8 ± 8.9 (their Table 1; 3 training seeds): our episode-end Reach 37.5% [18.5–61.4] is inside that CI.

task	arm	success	final state-dist	final ee-dist
mw-reach	L_2	6/16 (37.5%)	0.332	0.022
	grounded	8/16 (50.0%)	0.412	0.031
mw-push	L_2	0/16	0.624	0.038
	grounded	0/16	0.527	0.028
mw-pick-place	L_2	0/16	0.578	0.035
	grounded	0/16	0.496	0.021

Key readings (CLAIMS 2.1–2.7): (i) the harness reproduces the published baseline where BB is low; (ii) where BB is high both planners fail with the predicted signature — final ee 2–4 cm, object unmoved

at state-dist $\sim 0.5\text{--}0.6$ — matching the upstream paper’s qualitative report that the model “hallucinates grasping the object” and the field-wide pattern (V-JEPA-2-AC requires three hand-crafted sub-goal images for pick-and-place; Octo grasp 0–20%); (iii) the model-side fix improves the cost surface: paired Δ (final state-dist), pooled contact ($n=32$ pairs): $+0.089$ [bootstrap $+0.022, +0.162$]; push $+0.097$ [$-0.014, +0.221$], pick-place $+0.081$ [$+0.007, +0.160$]; no-harm on reach; (iv) no successes flip: BB repair is necessary but not sufficient — CEM rarely samples a contact-creating sequence for the grounded term to score (contact-aware action proposal is planner-side future work).

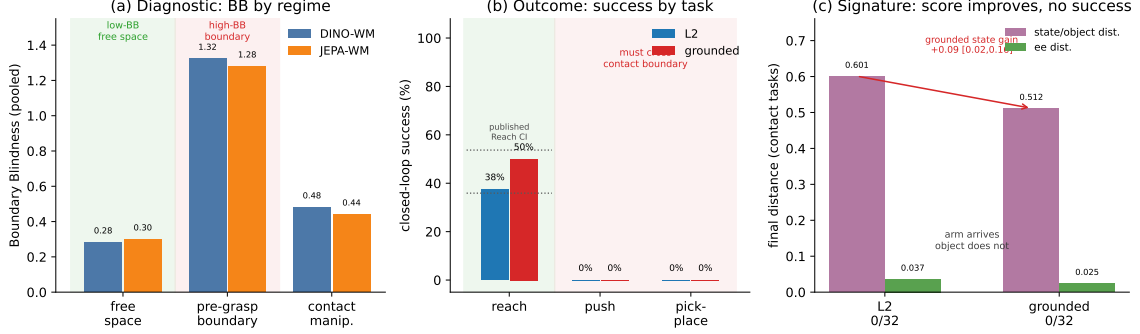


Figure 3: Boundary Blindness predicts closed-loop planning outcome at the regime level, not by a dense pointwise correlation. (a) BB is low in free space and high at the pre-grasp boundary on both Metaworld model families. (b) The low-BB free-space task reproduces the published Reach baseline (37.5%/50.0%, inside the CI), while the high-BB contact tasks fail (0%). (c) The contact failures have the BB signature: the end-effector is close but the object/state is not; grounded dynamics improves final state distance ($+0.089$, CI [$+0.022, +0.162$]) without flipping success.

5 Localizing the failure: a fix ladder on frozen trunks

5.1 What does not work (quantified nulls)

Mixture predictor heads (soft-NLL / WTA / hard-EM / supervised mode assignment; $K \in \{2, 3\}$; optional state-slice context): every variant beats its $K=1$ control on NLL; none moves BB_{boundary} . Two measured causes: (a) the bifurcation spans only $9.9 L_2$ units against a 106 median residual — the latent geometry under-weights the boundary subspace by an order of magnitude; (b) the boundary’s *action*-dependence is unlearnable from expert-only data at the head level (π action-flip rate ≈ 0). **Metric re-weighting** (probe-subspace φ -metric): redistributes BB without reducing it — one cannot expose a signal the predictor does not produce.

5.2 Where the information lives (probe chain)

V1: object state is decodable from the frozen latent (err 0.064 vs. sd 0.094). V2: it propagates through the predictor for *factual* actions (0.059). V3: the *counterfactual* action→object response is noise (spread corr $+0.035$). The broken piece is precisely the counterfactual action→object channel.

5.3 What works: a grounded dynamics channel

The diagnosis says the object’s action-dependence exists in the latent but is not exposed by the predictor. So rather than change the predictor or the metric, we add a small *auxiliary* channel that predicts the object outcome directly, supervised by the one signal the full-latent L_2 objective under-weights.

Architecture. A grounded dynamics head $h(z_t, a) \rightarrow \widehat{\Delta\text{obj}}$ reads the frozen latent token grid and the action. It pools the N latent tokens $z_t \in \mathbb{R}^{N \times D}$ by concatenating their mean and max, embeds the action with a one-layer MLP ϕ_a , and maps the concatenation to the object displacement with a small MLP:

$$h(z_t, a) = \text{MLP}([\text{mean}_n z_t \parallel \text{max}_n z_t \parallel \phi_a(a)]) \in \mathbb{R}^3. \quad (7)$$

At $\sim 0.5\text{M}$ parameters this is two orders of magnitude smaller than the frozen predictor (17–21M). The encoder, predictor, and action encoder are all frozen; only h trains, on the *cached* latents — no environment, no backprop through the world model (“force-free”).

Training. We regress object displacement with a plain MSE,

$$\mathcal{L}(h) = \mathbb{E}_{(z_t, a_t, s_t, s_{t+1})} \|h(z_t, a_t) - (\text{obj}(s_{t+1}) - \text{obj}(s_t))\|^2, \quad (8)$$

where $\text{obj}(\cdot)$ is the object slice of the Metaworld state (3-dim; on DROID, lacking object state, the label degrades to the full proprioceptive state difference — the transfer attempt that fails, Sec. 5 honest negatives). Three epochs of Adam at 10^{-3} on the cached transitions suffice.

Counterfactual gate. Before trusting h we verify it learned an action-dependent object response, not a per-state constant: on held-out boundary neighbourhoods we measure the correlation between the spread of h ’s predictions across the neighbourhood’s actions and the spread of the true outcomes; we accept only if this counterfactual-tracking correlation exceeds 0.2. The frozen predictor scores 0.035 (DINO); h scores 0.682 — it has recovered the channel.

Use at test time. h supplies a grounded read-out of the predicted object position. For the BB evaluation we substitute h ’s object-spread for the predictor’s latent spread in S^{model} . For planning we add a grounded term to the CEM cost: integrating h along the unrolled plan gives a predicted object trajectory whose endpoint is compared to the goal’s probed object position, weighted by β and normalised by a pooled scale s_g , on top of the upstream latent- L_2 term:

$$C(\text{plan}) = \underbrace{d(F^{(H)}(z_t, \mathbf{a}), z_g)}_{\text{upstream } L_2} + \beta \left\| \frac{1}{s_g} (\widehat{\text{obj}}_H - \text{obj}(z_g)) \right\|^2, \quad \widehat{\text{obj}}_H = g(z_t) + \sum_{t=1}^H h(F^{(t)}, a_t), \quad (9)$$

with g the object probe (the read-out used to initialise and to define the goal). This is the “grounded” planner arm of Sec. 4; note the squared-norm-over-scale normalisation (an earlier per-dimension MSE scaling made the latent term vanish and is disclosed as a corrected bug).

Table 5: The fix, on both model families (Metaworld; pooled, door-close excluded).

	DINO-WM	JEPA-WM
counterfactual tracking corr (frozen $\rightarrow +h$)	0.035 $\rightarrow$ 0.682	0.268* $\rightarrow$ 0.702
BB _{boundary} pre-grasp	1.323 $\rightarrow$ 0.660 (−50%)	1.280 $\rightarrow$ 0.620 (−52%)
boundary-vs-free gap	1.04 $\rightarrow$ 0.32	0.98 $\rightarrow$ 0.27

*probe-only ceiling on JEPA-WM.

The mixture-null + grounded-success pair revises the natural reading: the bottleneck is not missing counterfactual *data* (cross-sample neighbourhood variation suffices) but the *training target* — full-latent L_2 buries the object’s action-dependence.

5.4 Honest negatives

The recipe does not transfer to DROID’s noisy whole-state proxy label at 2.1k transitions (val MSE 0.865; rank correlation +0.708 but spread magnitude collapsed; BB unchanged): the principle needs a learnable grounding label, which real-robot datasets currently lack. Open-loop Action Error is no-harm/no-gain (the metric rewards full-arm expert mimicry); a scale-bug run is disclosed in the artifact set.

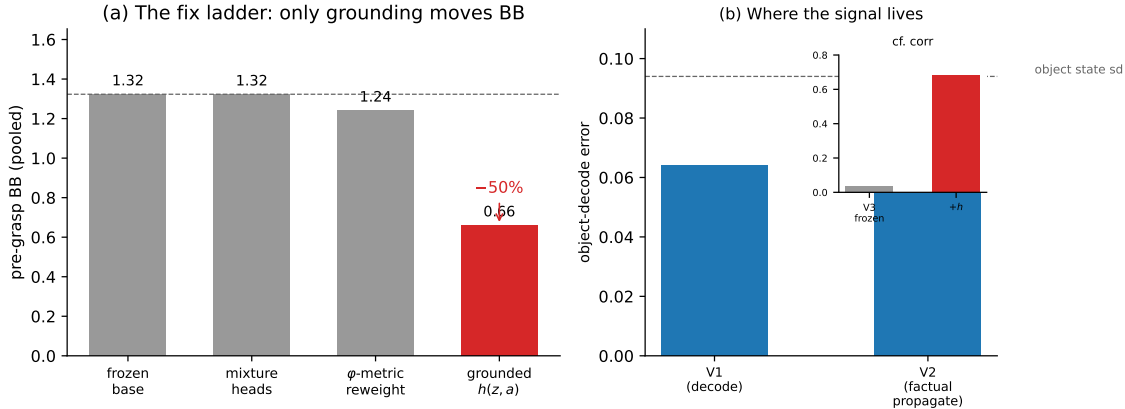


Figure 4: The fix ladder. (a) Pooled pre-grasp BB at each rung: mixture heads (1.32) and φ -metric reweighting (1.24) leave BB essentially at the frozen-base level (1.32); only the grounded dynamics channel moves it (0.66, -50%). (b) The probe chain: object state is decodable from the latent (V1) and propagated for factual actions (V2), but the counterfactual action→object response is noise (V3, corr 0.035) until the grounded channel restores it (corr 0.682, inset).

6 Discussion, scope, and limitations

What the closed-loop result does and does not claim. The 0/16 on push and pick-and-place is a property of the frozen baseline that our diagnosis predicts, not a headline success number we are advertising. Its evidential value rests on the reach result: at the same protocol, on the same checkpoint, judging success at the end of the episode (the upstream convention), the L_2 planner reaches 37.5% [18.5–61.4] and the grounded planner 50.0% — both inside the published 44.8 ± 8.9 CI — so the harness is healthy and the contact failure is the model’s. We verified the rendered observations land on the training manifold (one-step prediction error $1.4\times$ the dataset level, latent nearest-neighbour ratio 0.97) and that the physics is identical to data generation (action replay end-effector error 1.5 mm); the entire gap was visual-domain reproduction, which we fixed (Appendix A). One caveat tempers the reach comparison: we evaluate one released checkpoint over 16 episodes where the upstream paper averages three training seeds over 96, so we read the reach number as “reproduces, inside the reported CI” rather than a seed-matched replication.

Necessary but not sufficient. The grounded fix is a genuine but partial result. It repairs the model-side quantity — counterfactual tracking and Boundary Blindness — and that repair reaches the planner as a CI-supported improvement in final object distance on contact tasks (+0.089 [+0.022, +0.162]), with no harm in free space. It flips no successes, because the cross-entropy planner rarely *samples* a contact-creating action sequence for the grounded term to score. Boundary Blindness is thus necessary but not sufficient for contact planning: a boundary-aware *action proposal* — automating the hand-crafted sub-goals that Assran et al. (2025) supply manually, e.g. by seeding the planner with contact-directed primitives derived from the probed object position — is the natural next step, and is deliberately a planner-side contribution we leave to future work rather than fold into this model-centric paper.

Scope of the grounding recipe. Metaworld is the boundary *proof*: it is the dataset whose simulator state lets us define, supervise, and measure the boundary against ground truth. The boundary label is an object-displacement proxy (the released dataset carries no MuJoCo contact ground truth); we exclude the articulated-object task mw-door-close where the proxy misbehaves. DROID is a transfer check only — its proprioception is 7-dim pose-plus-gripper with no force, torque, or joint channel, so BB there uses the weaker $\|\Delta z\|$ proxy and the grounded *recipe* does not transfer at the 2.1k-transition scale we have (the label is not learnable; the principle is untouched). We make no tactile or force grounding claim on real robots.

Model coverage. Directly fixed and planned: DINO-WM and JEPA-WM on Metaworld, the only dataset with simulator state precise enough to supervise and measure the boundary against ground truth. Directly scaled diagnostically: DINO-WM (DINOv2), JEPA-WM/DINOv3, V-JEPA-2-AC, and V-JEPA-2-AC-OSS on DROID, all at the hard-negative chance floor with positive BB under the $\|\Delta z\|$ transfer proxy. Externally corroborated planning wall: V-JEPA-2-AC’s own published reliance on hand-crafted sub-goals for pick-and-place. The DROID/V-JEPA2 leg is therefore a diagnostic scaling result, not a closed-loop or grounded-fix claim.

7 Conclusion

Action-conditioned JEPA world models do not fail at contact for want of capacity, a better metric, or more counterfactual data — they fail because the prediction target, a full-latent L_2 to the next frame, averages over the outcome bifurcations that contact creates. We named this failure Boundary Blindness and measured it on frozen checkpoints, where it concentrates at the pre-grasp boundary on two model families and transfers to real-robot data; we tied it to planning, reproducing free-space success within the published baseline CI while contact tasks fail with the predicted signature of an arm that arrives at a motionless object; and we localized the cause with a fix ladder whose only working rung is a 0.5M-parameter grounded dynamics channel $h(z, a) \rightarrow \Delta \text{object}$ that, with the encoder, predictor, and data all frozen, restores counterfactual tracking ($0.03 \rightarrow 0.68$), halves boundary Boundary Blindness, and moves the closed-loop planning cost in the right direction. The remaining gap to task success is a planning problem, not a representation one — a clean separation that the diagnostic makes visible.

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A Reproduction integrity

Closed-loop sim evaluation is fragile, and getting the baseline wrong would invalidate the entire C2 argument. We document the three environment-side bugs we found and the checks that confirm the fixes, all of which are model-agnostic (they concern how the simulator is rendered, not the world model).

(1) Wrong camera at the wrong resolution. Metaworld constructs its offscreen renderer in the environment constructor, so assigning the camera name and image size afterwards is silently ignored: every initial closed-loop run rendered the default free camera at 480px instead of the `corner2` view at 224px the checkpoints were trained on. We mirror the upstream wrapper and re-create the renderer explicitly. Symptom before the fix: one-step prediction error $8.5\times$ the dataset level and rendered latents $3.3\times$ off the training manifold.

(2) Missing vertical flip. The released training frames are vertically flipped relative to today’s `corner2` render. We established this by pixel-calibrating against dataset initial frames over a grid of historically plausible camera configurations: the flipped wrapper-camera render scores mean-squared error 71.6 against the dataset, versus ≥ 3000 for every unflipped candidate. The flip is part of the data-generation pipeline, so the planner must feed the encoder flipped frames; an earlier removal of this flip had been judged on default-camera frames (the wrong camera, hence the wrong conclusion).

(3) Goal-frame convention. The success flag fires when the end-effector enters a 5 cm radius, but the scripted expert keeps refining to ~ 1 cm. Taking the goal frame at first-success rather than at the expert’s final frame parked the planner just outside the success radius — producing the systematic “arrives but scores zero” pattern (final ee 2–4 cm, success=0) that initially looked like a model failure.

Verification. After all three fixes, rendered transitions have one-step prediction error $1.4\times$ the dataset level and latent nearest-neighbour ratio 0.97 (i.e. on the manifold), and replaying a dataset trajectory’s raw actions in the live environment reproduces its proprioceptive trajectory to a median end-effector error of 1.5 mm — confirming that only the visuals had drifted, never the physics. The reach baseline then recovers from near-zero (broken renderer) to the episode-end 6/16 (37.5%) reported in Sec. 4, inside the published CI. The probe scripts that produce these numbers are released with the code.

B AUG and ECS: the raw-gap corroboration

The main text leads CounterfactualBench with CRA because it is a ranking and so comparable across models with different latent scales (Sec. 3.1). For completeness we report the two raw-gap metrics, AUG and ECS, here. Both are measured on the hard nearest-neighbour distractor family (the discriminative regime; *opposite* distractors are saturated) and restricted to effect-bearing transitions ($\|z_{t+1} - z_t\| > \tau$, per-model median); ECS is additionally the effect-mask form. Cells are n_{eff} -weighted pools. A dash means the regime had no effect-bearing transitions on that dataset ($n_{\text{eff}} = 0$: DROID free-space, Metaworld gripper) so the metric is undefined there.

The pattern matches the main result from a different angle: the effect-conditioned action gap is statistically indistinguishable from zero on the real-robot data across the full $45\times$ scale range, and small (and, at pre-grasp, no larger than free space) in simulation. Because AUG/ECS are not cross-model comparable they cannot themselves carry the scaling claim — that is CRA’s job — but they confirm that the chance-level CRA is driven by genuine action-ignoring rather than a quirk of the ranking. Source: `results/{droid,metaworld}_diagnostic.csv`, `hard_nn` rows.

C Full per-task BB tables and ablations

[TODO: per-task CSV dumps; mixture-K, boundary-head on/off, state-slice ablations]

Table 6: AUG / ECS on the hard nearest-neighbour family, per regime. Both collapse toward zero everywhere — feeding the *realised* action moves the prediction no closer to the next latent than a permuted action ($\text{AUG} \approx 0$), and the effect-conditioned sensitivity is at noise (ECS within ± 0.01 of 0 on DROID, and small but positive on Metaworld). This corroborates the CRA-at-chance result: the models ignore the action where it matters. **These are raw latent gaps and are *not* comparable across models** (e.g. DINO-WM’s larger DROID values reflect a larger latent scale, not more action use; the cross-model statement lives in CRA, Table 1, and the scaling curve, Table 3).

model	free space	pre-grasp	gripper	contact
<i>DROID</i> (real robot; AUG / ECS)				
DINO-WM (22M)	0.038/−	0.011/0.0105	0.038/0.0645	0.070/0.0094
JEPA-WM (300M)	0.000/−	0.000/0.0002	0.001/0.0004	0.001/−0.0011
V-JEPA-2-AC (1B)	0.006/−	0.002/0.0016	0.003/0.0026	0.003/−0.0059
V-JEPA-2-AC-OSS (1B)	0.004/−	0.001/0.0010	0.001/0.0005	0.003/−0.0030
<i>Metaworld</i> (sim; AUG / ECS; pooled over 12 tasks)				
DINO-WM (22M)	0.088/0.1007	0.051/0.0997	−/−	0.081/0.0893
JEPA-WM (300M)	0.089/0.1020	0.058/0.1114	−/−	0.095/0.1052